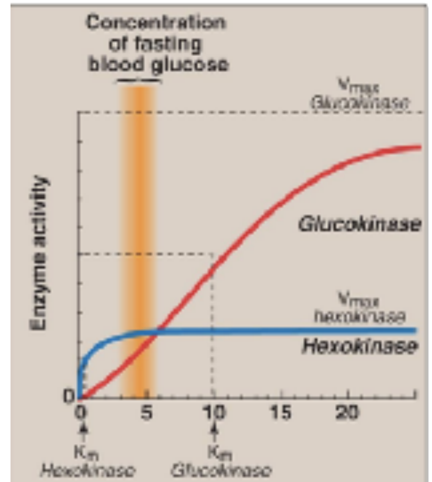
1. GLUT transporters (facilitated diffusion of glucose from blood into cells)
    1. GLUT1: RBCs and neurons/brain
    2. GLUT2: Liver
    3. GLUT3: neurons/brain
    4. GLUT4: muscle/fat (insulin dependent!)
2. Glucokinase vs Hexokinase

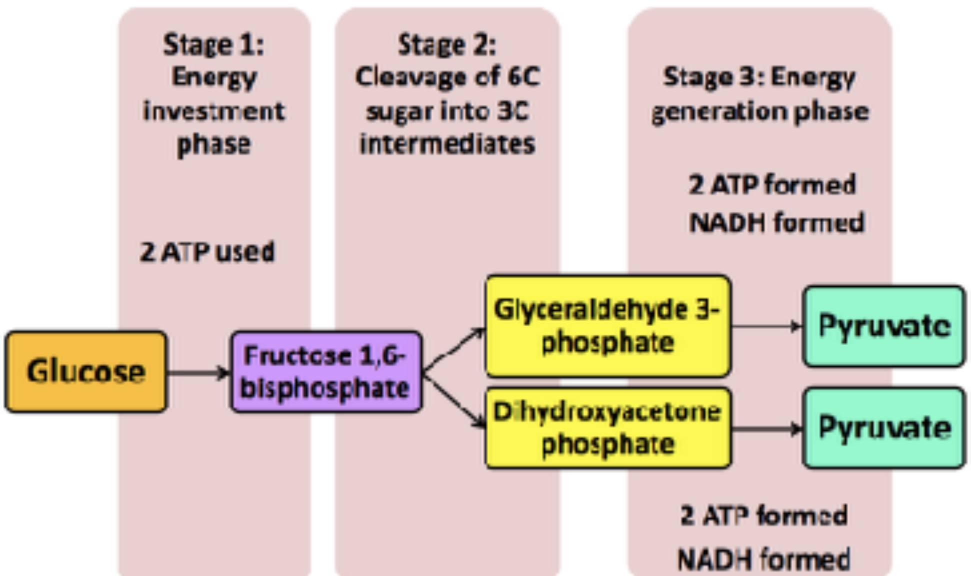
    Glucokinase: Liver/beta cells of pancreas. High Km glucose sensor!)

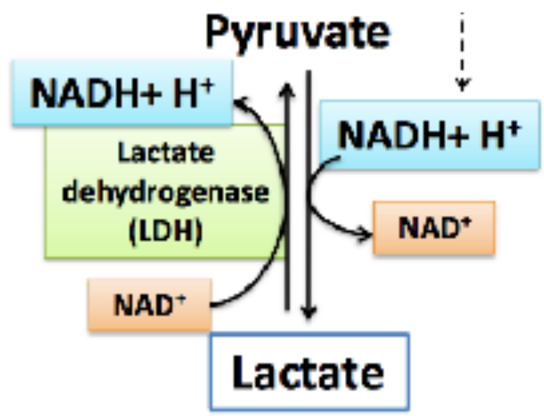
    Hexokinase: Other tissue. Low Km

    Glucokinase mutation: Hypoglycemia, MODY-2



3. Glycolysis and PPP both in cytosol: work even in cells w/o mitochondria





4. Pyruvate/Lactate Balance: Determined by NADH/NAD+ ratio

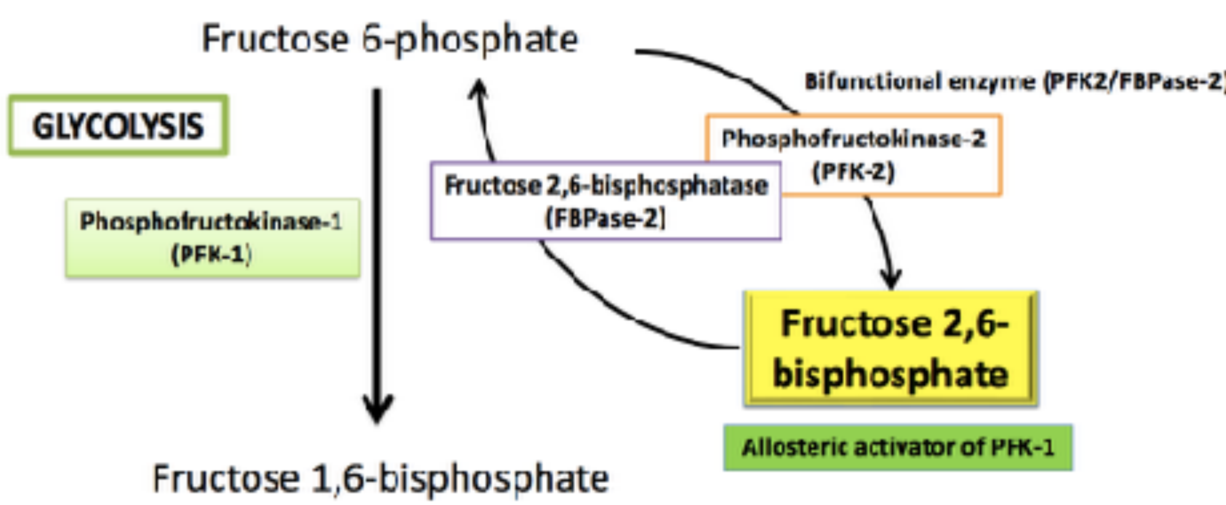
    -When you have lots of NADH, you will form lactate —> Lactic acid

    -When NADH is depleted, you are stuck in pyruvate —> PDH —> Acetyl CoA —> TCA

    -Ox. Phosphorylation uses up NADH. So in cells with a lot of mitochondria and a lot of available oxygen (thus lots of ox. phosphorylation occurring), NADH levels are kept low. Low NADH levels means Pyruvate predominates —> lots of Pyruvate converted to Acetyl CoA —> TCA (also makes NADH)

    *Extra Lactate can be sent to liver —> Gluconeogenesis (Cori Cycle)

5. Too much Lactate —> Lactic Acidosis (metabolic acidosis)
   a. Binge drinking (increases NADH/NAD+ ratio)
   b. PDH Deficiency- Leigh's Disease (Pyruvate builds up, some converted back to Lactate)
   c. Thiamine Deficiency- PDH can't function
   d. Problem with Gluconeogenesis/Cori Cycle (eg Von Gierke)- can't transport/use the lactate
   e. Anything that decreases blood supply/oxygen (decreases ox. phosphorylation
6. Effect of Arsenic: Inhibit Glyceraldehyde 3-Phosphate Dehydrogenase
   -buildup of Glyceraldehyde 3P
   -Low NADH (this is the step were NADH would be made!)
   -Low ATP formation
7. 3 Regulated Reactions of Glycolysis
   a. Glucokinase (regulated by glucose sensor function)
   b. PFK-1 (regulated by bifunctional enzyme)
   c. Pyruvate Kinase (Feedforward by Fructose 1,6-bisphosphate)
8. Regulation of Glycolysis: The Bifunctional Enzyme
   a. Low blood glucose —> glucagon —> enzyme phosphorylated —> FBPase-2 —> INHIBITION
   b. High blood glucose —> insulin —> enzyme Dephosphorylated —> PFK2 (Phosphofructokinase-2) —> Make more 2,6 Bisphosphate —> ACTIVATE GLYCOLYSIS



   -Feedforward activated (high conc of Fructose 1,6-bisphosphate)

- Inhibited when Alanine is high
    - Excess Pyruvate is converted to Alanine. High Ala means you already have plenty of Pyruvate. Don't need more glycolysis

    *Pyruvate Kinase is inhibited when phosphorylated!